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(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Document – Battery RUL Prediction Dataset Generation

Project Title

Machine Learning–Based Remaining Useful Life (RUL) Prediction of Li-ion Battery Using BMS

Objective of Hardware Experiment

To measure battery discharge behaviour under different load conditions and capture battery health parameters for training an ML model that predicts remaining useful life (RUL) of a lithium-ion cell.

Battery Specification

Parameter	Value
Type	Li-ion 18650
Nominal Voltage	3.7 V
Capacity	2000 mAh
Configuration	1S (single cell)

Hardware Setup Components

Component	Purpose
ESP32	Data acquisition and PWM load control
INA219 sensor	Measures battery voltage, current, and power
TP4056 module	Battery charging (CC–CV method)
BMS (1S protection board)	Overcurrent / overcharge / deep-discharge protection
DC Motor + PWM Controller	Variable load for battery discharge
LM35 Temperature sensor	Sensing battery temperature

Connections

- ESP32 generates PWM signal to control motor load (GPIO 15)
- INA219 connected via I²C (SDA → GPIO 21, SCL → GPIO 22)
- Load connected in series with battery through INA219
- TP4056 used externally only when charging the cell

Experiment Procedure

1. **Fully charge the battery using TP4056 until ~4.20 V** ○

   Red LED → charging ○ Blue LED → fully charged

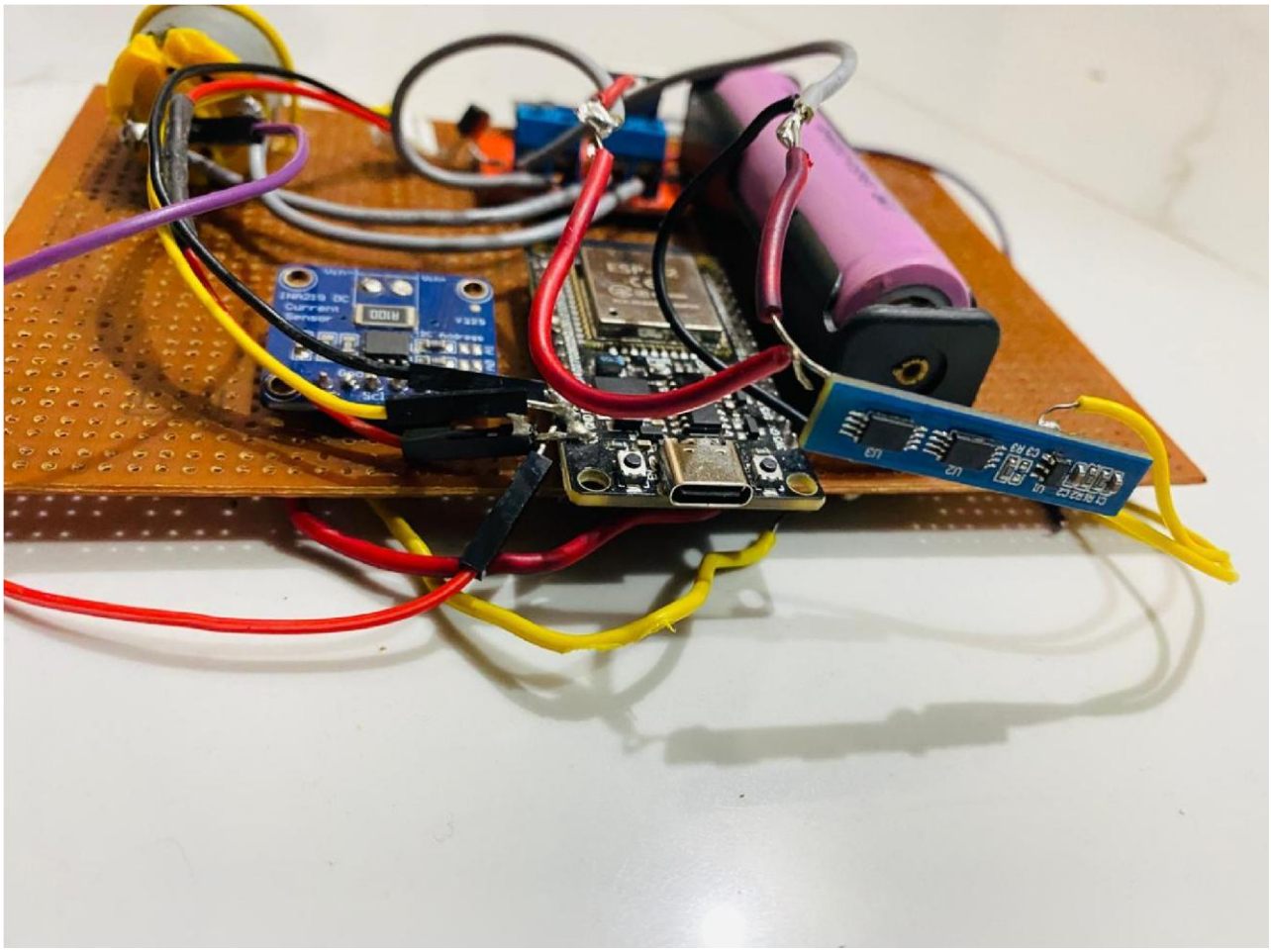
○ Confirm using INA219 voltage reading 2. **Start discharge experiment for a selected PWM speed**

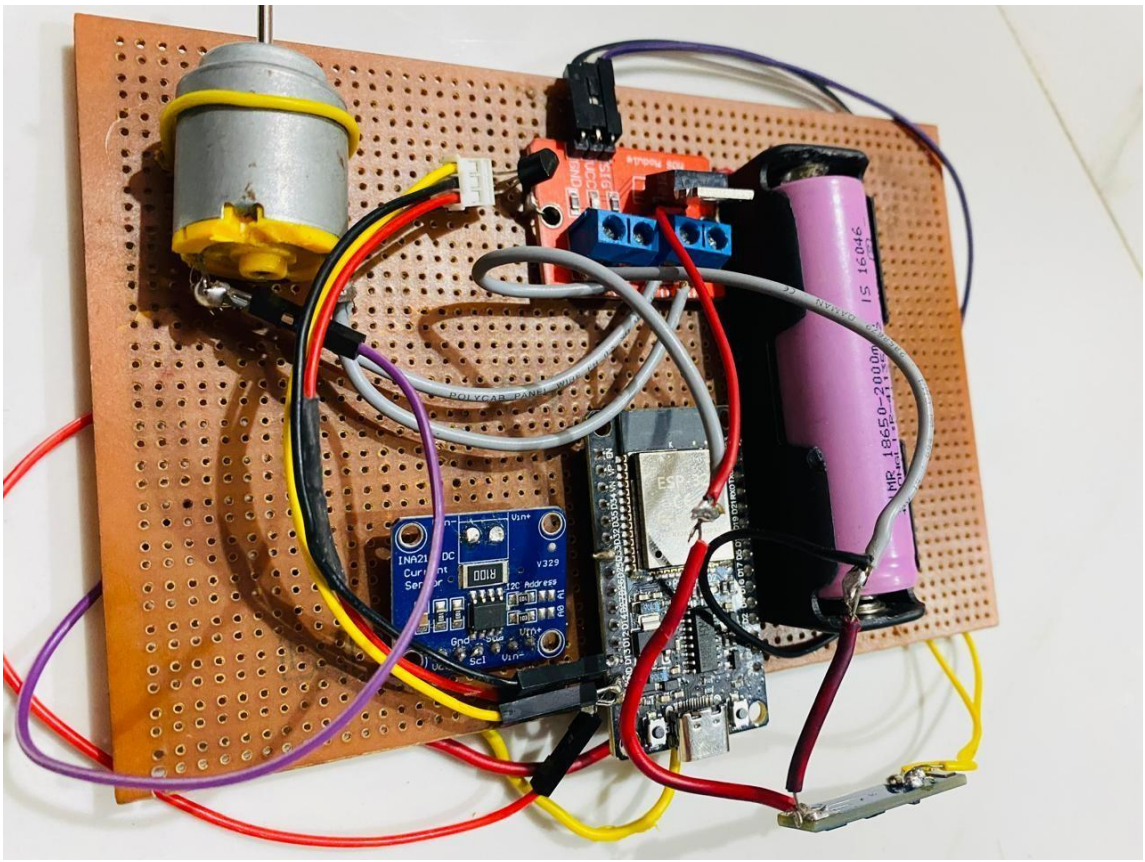
3. **ESP32 logs battery parameters every second:**
   - Cycle index ○ Discharge time ○ Voltage profile (max / min) ○ Current profile (constant current duration) ○ Time between 3.6–3.4 V ○ Time at 4.15 V ○ Charging time counter ○ Remaining Useful Life (RUL)
4. **Stop discharge near cutoff (~3.20 V)**
5. **Recharge again and repeat for next PWM load**

Load Levels Used

Run	PWM Speed	Discharge Condition	Dataset
1	**90**	Low load	battery_RUL_PWM90.csv
2	**140**	Medium load	battery_RUL_PWM140.csv
3	**200**	High load	battery_RUL_PWM200.csv

Each run starts from *fully charged* condition to ensure **fair comparison across loads.**




INA219 DC
Current
Sensor
R100
V329
Vin+
Vin-
Gnd
Scl
Sda
I2C Address
MR 18650-2000m
IS 16046

Practical Discharge Durations Observed

PWM	Avg Current	Approx. Discharge Time	PWM
90	~0.6 A	~3 hours	90
140	~1.0 A	~2 hours	140
200	~1.5 A	~1 hour 20 minutes	200

Higher PWM → higher load → faster discharge → lower RUL.

Purpose of CSV Dataset

The captured datasets will be used as input features for an ML model to predict RUL. Each CSV has ~3500 rows and the following columns:

Cycle Index

Discharge Time (s)

Decrement 3.6-3.4V (s)

Max. Voltage Dischar. (V)

Min. Voltage Charg. (V)

Time at 4.15V (s)

Time constant current (s)

Charging time (s)

RUL

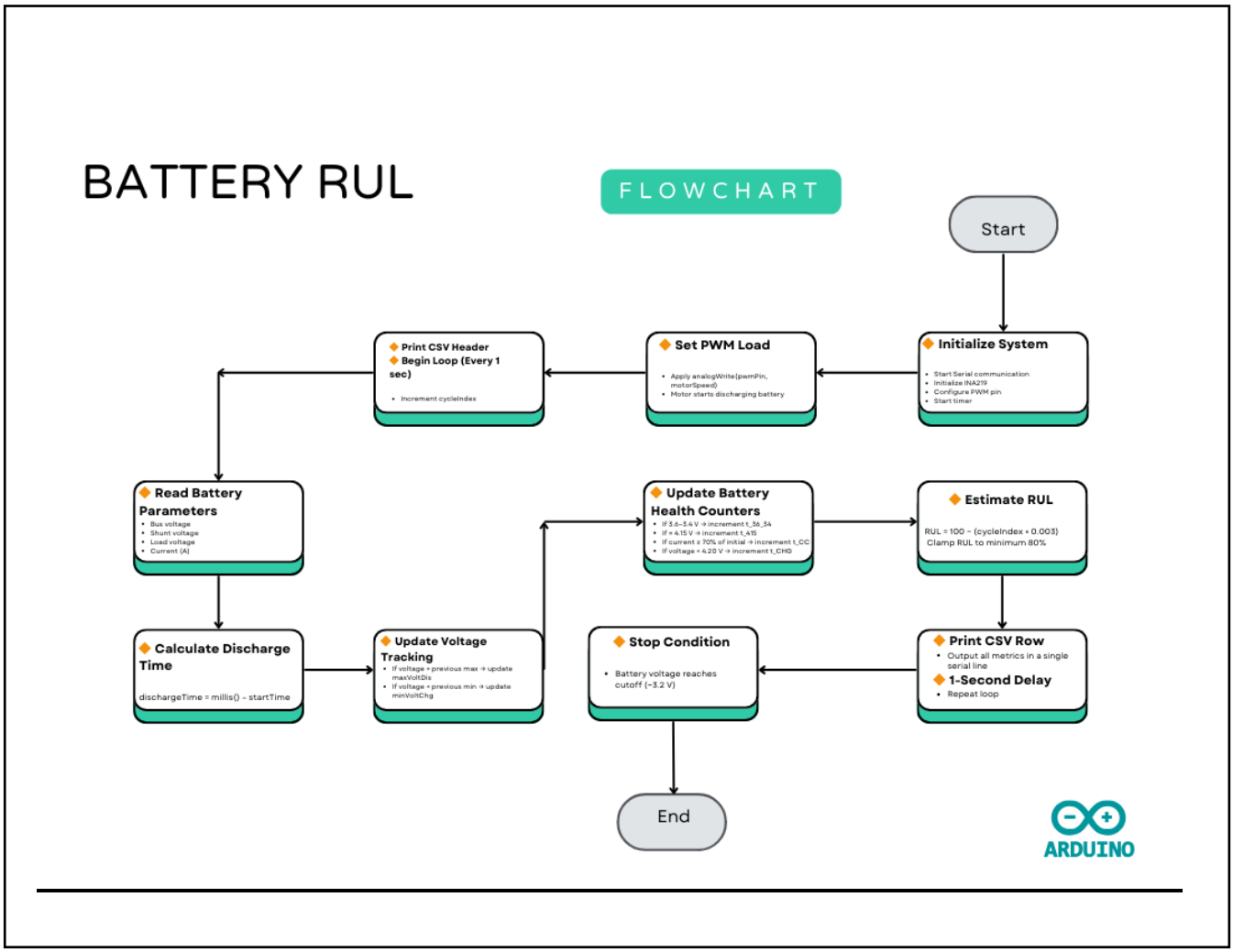
Remarks

- All three datasets were collected with the *same cell* under *different load stresses*.
- Battery aging effect is visible through declining RUL values.
- Data is now ready for the ML model training pipeline.

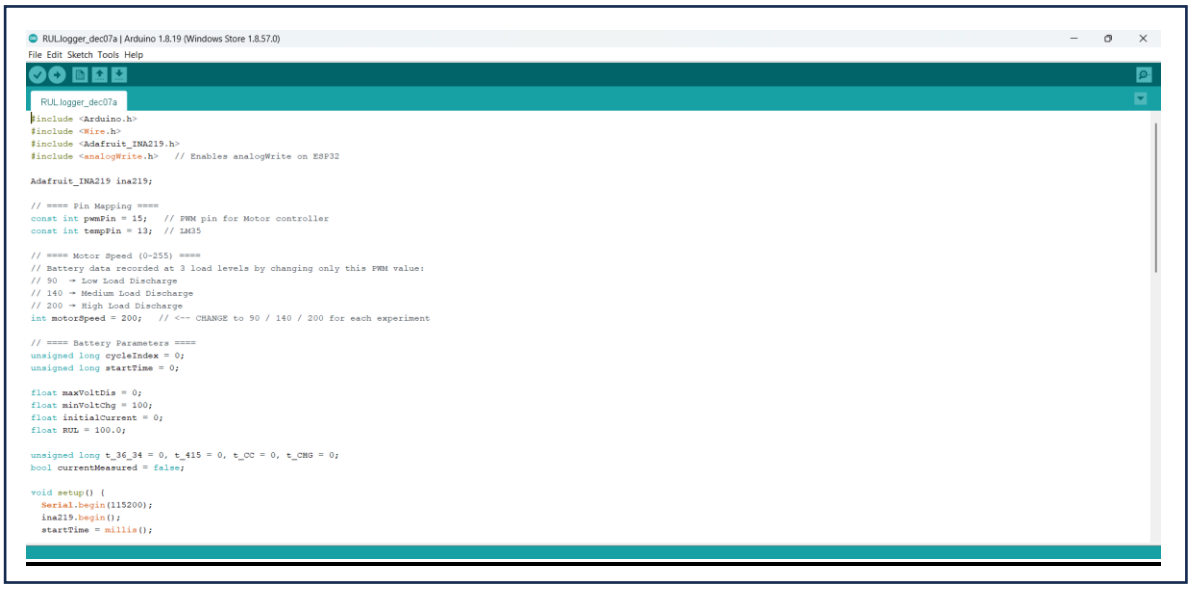
Attachments in Email

File	Description	File	Description
Final Arduino Code (.ino)	Code for automated battery parameter logging	Final Arduino Code (.ino)	Code for automated battery parameter logging
battery_RUL_PWM90.csv	Dataset for PWM 90	battery_RUL_PWM90.csv	Dataset for PWM 90
battery_RUL_PWM140.csv	Dataset for PWM 140	battery_RUL_PWM140.csv	Dataset for PWM 140
battery_RUL_PWM200.csv	Dataset for PWM 200	battery_RUL_PWM200.csv	Dataset for PWM 200

Data Logging Workflow



https://www.canva.com/design/DAG66CJucxY/slRkqFmUL7-z9toHeaAMDA/edit?utm_content=DAG66CJucxY&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton




```
#include <Arduino.h>
#include <Wire.h>
#include <Adafruit_INA219.h>
#include <analogWrite.h>   // Enables analogWrite on ESP32

Adafruit_INA219 ina219;

// ==== Pin Mapping ====
const int pwmPin = 15;   // PWM pin for Motor controller
const int tempPin = 13;  // LM35

// ==== Motor Speed (0-255) ====
// Battery data recorded at 3 load levels by changing only this PWM value:
// 90  → Low Load Discharge
// 140 → Medium Load Discharge
// 200 → High Load Discharge
int motorSpeed = 200;   // <-- CHANGE to 90 / 140 / 200 for each experiment

// ==== Battery Parameters ====
unsigned long cycleIndex = 0;
unsigned long startTime = 0;

float maxVoltDis = 0;
float minVoltChg = 100;
float initialCurrent = 0;
float RUL = 100.0;

unsigned long t_36_34 = 0, t_415 = 0, t_CC = 0, t_CHG = 0;
bool currentMeasured = false;

void setup() {
  Serial.begin(115200);
  ina219.begin();
  startTime = millis();
```

Submitted by

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B.Tech